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RV COLLEGE OF ENGINEERING®
Autonomous Institution affiliated to VTU
VI Semester B. E. Examinations
Department of Information Science and Engineering
Model Question Paper
CRYPTOGRAPHY AND NETWORK SECURITY

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, one full question from 5 and 6, one full question from 7 and 8 & one full question from 9 and 10.

PART-A

Q.	SL No	Question	Marks
1.	1.1	List the two basic functions used in encryption algorithms?	02
	1.2	Differentiate between a block cipher and a stream cipher?	02
	1.3	Compare the efficiency of monoalphabetic cipher with that of a polyalphabetic cipher	02
	1.4	Mathematically define ECB mode of operation for Block ciphers.	02
	1.5	Illustrate cipher block chaining mode for both encryption and decryption.	02
	1.6	What is a meet-in-the-middle attack?	02
	1.7	Outline the two requirements for Key distribution under symmetric encryption.	02
	1.8	Define the structure of a trap door one way function.	02
	1.9	Give two approaches to produce message authentication.	02
	1.10	Define any two ways in which PGP uses the concept of trust.	02

PART-B

2	a	A ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is "B," and the second most frequent letter of the ciphertext is "U." Break this code.	06
	b	Answer the following with respect to Playfair cipher i) How many possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2. ii) Consider a scenario where some Playfair keys produce the same encryption results. How many effectively unique keys does the Playfair cipher have?	06
	c	Using the Vigenère cipher, encrypt the word "explanation" using the key <i>leg</i> .	04
3	a	Design Feistel Cipher Structure for a General n -bit to n -bit Block Substitution using $n=4$	06
	b	Discuss strength of DES with respect to: i) Use of 56 bit key ii) Nature of DES algorithm	10

OR			
4	a	Can we perform both encryption and decryption operations in parallel on multiple blocks of plaintext in CBC mode? Justify your answer.	08
	b	If a bit error occurs in the transmission of a ciphertext character in 8-bit CFB mode, how far does the error propagate? Justify your answer with a CFB design.	08
OR			
5	a	With the help of an illustration explain public key cryptosystem for Authentication	06
	b	In an RSA system, the public key of a given user is $e = 31$, $n = 3599$. What is the private key of this user? <i>Use the following steps to solve the problem.</i> i) Use trial-and-error to determine p and q ; ii) Use extended Euclidean algorithm to find the multiplicative inverse of 31 modulo $\phi(n)$.	10
OR			
6	a	In a public-key system using RSA, you intercept the ciphertext $C = 10$ sent to a user whose public key is $e = 5$, $n = 35$. What is the plaintext M ?	06
	b	Discuss requirements of public key cryptography.	06
	c	Outline any four possible ways to attack RSA algorithm.	04
OR			
7	a	Determine the changes required in HMAC in order to replace one underlying hash function with another.	06
	b	"It is necessary to recover the secret key in order to attack a MAC algorithm". Do you agree with this. Justify your answer with appropriate example.	06
	c	Differentiate between message authentication code and one way hash function	04
OR			
8	a	The data authentication algorithm can be defined as using the cipher block chaining (CBC) mode of operation of DES with an initialization vector of zero. Show that the same result can be produced using the cipher feedback mode.	08
	b	When a combination of symmetric encryption and an error control code is used for message authentication, determine in what order must the two functions be performed with appropriate explanation for the same.	08
OR			
9	a	With respect to SSL explain the following with illustrations: i) Connection and Session ii) Protocol Stack	08
	b	Tabulate any 4 PGP services along with its description, Algorithm used and applications	08
OR			
10	a	Discuss the major security services provided by AH and ESP, respectively.	08
	b	Enumerate any four general means to authenticate User identity and give example for each.	08

..... Q.P.Code

Course Code: IS362IA

Course Title: CRYPTOGRAPHY AND NETWORK SECURITY

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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	2	2	2	1	1	2	1	1	1
COs	1	1	2	2	2	1	2	1	1	1
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	3	3		3	a	3	3
	b	3	3			b	2	3
	c	3	3			c		
	d					d		
4	a	2	3		5	a	2	3
	b	3	3			b	3	3
	c					c		
	d					d		
6	a	3	2		7	a	1	1
	b	1	1			b	3	1
	c	1	1			c	3	1
	d					d		
8	a	3	2		9	a	2	4
	b	3	2			b	2	2
	c					c		
	d					d		
10	a	1	4		11	a		
	b	2	2			b		
	c					c		
	d					d		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name: